

JÉRÔME GARNIER-BRUN

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French citizen, born March 1998.

EXPERIENCE

Università Bocconi, Milan

June 2025 – current

Marie Skłodowska-Curie Research Fellow

Marie Skłodowska-Curie Action funding for project SPIRAL (Structured Physics-Inspired Representations and dAta models for efficient Learning), carried out under the guidance of Prof. M. Mézard.

Università Bocconi, Milan

Jan. 2024 – June 2025

Postdoctoral research fellow

Grant holder in the Theoretical Physics of Matter sector, working with Prof. M. Mézard (Dept. of Computing Sciences) on the role of the structure of data in learning and Prof. C. Tebaldi (Dept. of Finance) on networks of firms.

Kyoto University, Kyoto

July 2022 – Oct. 2022

Visiting PhD student

JSPS fellowship for research on accelerating slow dynamics using nonequilibrium driving with Prof. S.-i. Sasa.

ENSAE and École polytechnique, Paris

Oct. 2020 – June 2022

Teaching assistant

Supervision of tutorials in socio/econophysics and simulations in Python (84h total, 100% positive student ratings).

EDUCATION

École polytechnique, Paris

Oct. 2020 – Dec. 2023

PhD, Theoretical Physics

Thesis on the out-of-equilibrium statistical physics of complex and disordered systems, with a focus on socioeconomic-inspired models [[link](#)]. Supervised by Profs. J.-P. Bouchaud and M. Benzaquen at the [EconophysiX lab](#).

Sorbonne Université, Paris

Sept. 2019 – June 2020

Master of Physics, Physics of Complex Systems

Highest honours

Imperial College, London

Oct. 2015 – July 2019

Master of Engineering, Aeronautical Engineering

Highest honours

PUBLICATIONS

11. **G.-B. J.**, Biggio L., Beltrame D., Mézard M. & Saglietti L. (2026). “Biased Generalization in Generative Diffusion.” *Accepted at the 43rd International Conference on Machine Learning (ICML 2026)* (**spotlight paper**) [[arXiv](#)]
10. Ariosto S., **G.-B. J.**, Straziota D. & Saglietti L. (2025). “Replication and Information Extraction in a Minimal Agent-Environment Model.” *arXiv preprint*, arXiv:2509.23212 (under review at *Phys. Rev. Lett.*) [[arXiv](#)]
9. **G.-B. J.**, Zakine R. & Benzaquen M. (2025). “Hydrodynamics of Cooperation and Self-Interest in a Two-Population Occupation Model.” *Phys. Rev. Lett.* **135**, 107402 (**Editors’ Suggestion**) [[journal](#), [arXiv](#)], also featured in an École polytechnique press release [[link](#)]

8. **G.-B. J.**, Biggio L., Mézard M. & Saglietti L. (2025). “Early-stopping Too Late? Traces of Memorization Before Overfitting in Generative Diffusion.” *The Impact of Memorization on Trustworthy Foundation Models: ICML 2025 Workshop* [[OpenReview](#)]
7. **G.-B. J.**, Mézard M., Moscato E. & Saglietti L. (2025). “How Transformers Learn Structured Data: Insights From Hierarchical Filtering.” *Proceedings of the 42nd International Conference on Machine Learning (ICML 2025)*, PMLR **267** [[OpenReview](#)]
6. **G.-B. J.**, Benzaquen M. & Bouchaud J.-P. (2024). “Unlearnable Games and ‘Satisficing’ Decisions: a Simple Model for a Complex World.” *Phys. Rev. X* **14**, 021039 [[OA journal](#)], also featured in the popular science magazine *New Scientist* [[link](#)]
5. Zakine R., **G.-B. J.**, Becharat A.-C. & Benzaquen M. (2024). “Socioeconomic agents as active matter in nonequilibrium Sakoda-Schelling models.” *Phys. Rev. E* **109**, 044310 [[journal](#), [arXiv](#)]
4. Dechant A., **G.-B. J.** & Sasa S.-I. (2023). “Thermodynamic Bounds on Correlation Times.” *Phys. Rev. Lett.* **131**, 167101 (**Editors’ Suggestion**) [[journal](#), [arXiv](#)]
3. **G.-B. J.**, Bouchaud J.-P. & Benzaquen M. (2023). “Bounded rationality and animal spirits: a fluctuation-response approach to Slutsky matrices.” *J. Phys. Complex.* **4**, 015004 [[OA journal](#)]
2. Alemany I., Rose J. N., **G.-B. J.**, Scott A. D. & Doorly D. J. (2022). “Random walk diffusion simulations in semi-permeable layered media with varying diffusivity.” *Sci. Rep.* **12**, 10759 [[OA journal](#)]
1. **G.-B. J.**, Benzaquen M., Ciliberti S. & Bouchaud J.-P. (2021). “A new spin on optimal portfolios and ecological equilibria.” *J. Stat. Mech.* **2021**, 093408 (featured in Highlights) [[OA journal](#)]

SEMINARS, SCHOOLS & CONFERENCES

- *Laboratoire Matière et Systèmes Complexes* (MSC), Université Paris Cité, 05/12/2025 — seminar
- *Laboratoire de Physique Théorique et Modèles Statistiques* (LPTMS), Université Paris-Saclay, 04/12/2025 — seminar
- *Laboratoire Interdisciplinaire de Physique* (LiPhy), Université Grenoble Alpes, 18/11/2025 — seminar
- *EconophysiX chair retreat*, Avignon, 13/10/2025 — seminar
- *Conference on Inference and AI* (Rockin’AI), Roccella Ionica, 02/09/2025 — contributed talk
- *Forty-Second International Conference on Machine Learning*, Vancouver, 13-19/07/2025 — poster
- *Learning with Machines, Physics and Minds summer school*, Beg Rohu, 02-13/06/2025
- *Towards a theory for typical-case algorithmic hardness winter school*, Les Houches, 27/01-07/02/2025
- *Young Seminar*, Società Italiana di Fisica Statistica (SIFS), 14/11/2024 — seminar
- *EconophysiX chair retreat*, Saint-Anne La Palud, 08/10/2024 — seminar
- *Conference on Inference and AI* (Rockin’AI), Roccella Ionica, 03/09/2024 — contributed talk
- *Glassy and Complex Systems summer school*, Cargèse, 08-19/07/2024 — poster
- *From Ecology to Economics and back workshop*, CFM & ENS, 13-14/05/2024 — **co-organizer**
- *Ergodicity Economics 2024*, Santa Fe Institute (remote), 19/02/2024 — contributed talk
- *Statistical Physics of Complex Systems summer school*, Beg Rohu, 05-16/06/2023 — poster
- *Recent advances in understanding artificial and biological neural networks winter school*, Les Houches, 20-24/02/2023 — poster
- *Yoshino lab.*, Osaka University, 06/10/2022 — seminar
- *Sasa lab.*, Kyoto University, 01/08/2022 — seminar
- *Laboratoire Gulliver theory group*, ESPCI Paris, 21/04/2022 — seminar

- *Nonequilibrium physics of living matter research group*, RIKEN Kobe (remote), 15/12/2021 — seminar
- *Conference on Complex Systems – Econophysics satellite*, Lyon, 27-28/10/2021 — poster
- *Fundamental Problems in Statistical Physics XV summer school*, Brunico, 12-24/07/2021
- *Glassy Systems and Inter-Disciplinary Applications summer school*, Cargèse, 28/06-08/07/2021 — poster

GRANTS & PRIZES

- Marie Skłodowska-Curie Action *Postdoctoral fellowship*, Sept. 2025 – Sept. 2027 [\[link\]](#)
- French Complex Systems Society *Best thesis prize* – finalist, Oct. 2024 [\[link\]](#)
- G-Research *Grant for PhD students*, Feb. 2023 [\[link\]](#)
- Japanese Society for the Promotion of Science (JSPS) *Summer fellowship*, Jul. – Oct. 2022 [\[link\]](#)
- Imperial College Department of Aeronautics *Best computational final year project*, June 2019
- Imperial College Faculty of Engineering *Dean’s list*, June 2018 & June 2019

REFERENCES

- Marc Mézard (Università Bocconi & Académie des Sciences): marc.mezard@unibocconi.it
- Jean-Philippe Bouchaud (CFM & Académie des Sciences): jean-philippe.bouchaud@cfm.com
- Michael Benzaquen (CNRS & École polytechnique): michael.benzaquen@polytechnique.edu

SUPERVISION

- Federica Kulka (Università Bocconi): Master thesis (2025), co-supervised with C. Lucibello
- Luca Alberto Polverino (Università Bocconi): Bachelor thesis “On the Statistics of Bitcoin Returns: Power-Law Tails, Volatility Dynamics, and Fractality” (2024), co-supervised with M. Mézard—presented at the *Complexity in Economics and Finance* StatPhys29 satellite
- Sayah El Hajji (ENS): Master thesis “Dynamics in low dimensional complex non-potential forces” (2021), co-supervised with J.-P. Bouchaud & M. Benzaquen
- Noé Beserman (ENS): Master thesis “A generalization of the Schelling model: abolishing frontiers” (2021), co-supervised with J.-P. Bouchaud & M. Benzaquen